

Final Project Paper

Introduction

Our group has decided to study the public health topic of healthcare access in Ohio. We set out to do this by examining vulnerable populations and the access to healthcare options that people have available to them. We have chosen to do this by first looking at the Ohio Department of Health's data on the Ohio Health Improvement Zones. From these zones, we were able to narrow that we wanted to focus on examining the Social Vulnerability Index (SVI) that is used to determine the OHIZ. SVI data is given by county, to see which populations are at greatest risk in Ohio, with the breakdown of all different kinds of SDOH affecting this value. We have also decided to use County-level health indicators to help figure out the healthcare access vulnerabilities in the counties. Doing research on this topic is of great importance to figure out where gaps in structural support are. We have chosen to find the vulnerable populations in Ohio, and then compared the overlap between the primary care physician rates per county to determine the resource availability of healthcare access.

Data description

The first data set includes the Social Vulnerability index by county. The Social Vulnerability index per County is found by compounding many individual SDOH factors. These SDOH factors are the non-medical ways in which people work, play and live in their environment and the effect it has on health outcomes. The data set compares many factors of Socioeconomic status, including income, unemployment rate, socioeconomic score and below poverty amounts. It also has factors of built environment, including transport ability, home crowding and types of homes/availability and usage of them. The data set also includes demographic factors such as minority status, disability score, English speaking ability, and minority and senior age population.

The second data set has many different county-level health indicators across all counties in Ohio. This all ranges in five different categories: Length of Life, Quality of Life, Health Infrastructure, Physical Environment ,and Social and Economic Factors. We chose to use this dataset primarily for the data it could provide on Health Care access. Even though it

compares many different aspects of county-wide health, this data set provided what we could use to examine the gaps in vulnerability and support.

Project Hypothesis

Our research question is: How does healthcare access differ across Ohio counties based on social vulnerability factors? What gaps exist between vulnerability and support? We expect that counties with higher social vulnerability will have less access to primary care physicians.

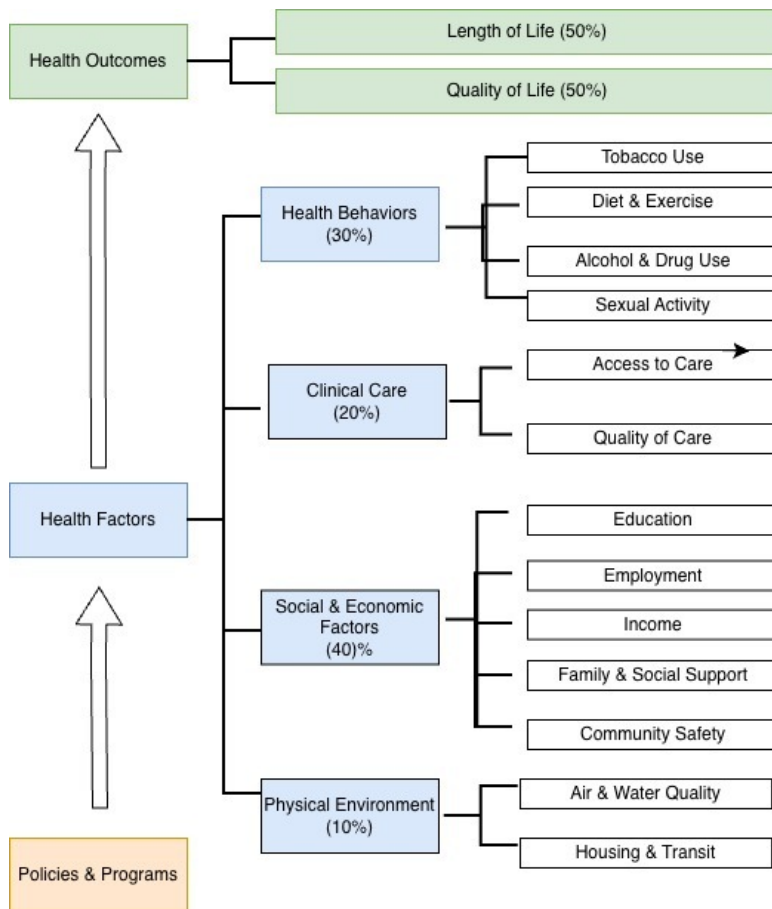
Methods

Our first dataset is from the Ohio Department of Health and is centered around Health Improvement Zones. This refers to the socioeconomic and demographic factors that can affect the resilience of individuals and the communities. The Ohio counties data that we are using has information on all 88 Counties with 16 different indicators grouped into four themes and the overall SVI. The indicators are grouped as follows:

Overall Vulnerability	Socioeconomic Status	Below 150% Poverty
		Unemployed
		Housing Cost Burden
		No HS Diploma
		No Health Insurance
	Household Characteristics	Aged 65+
		Aged 17 & Younger
		Civilian with a Disability
		Single-Parent Household
		English Language Proficiency
	Racial & Ethnic Minority Status	Hispanic or Latino, Black or African American, Two or More races, Other Races, etc
	Housing Type & Transportation	Multi-Unit Structures
		Mobile Homes
		Crowding
		No Vehicle
		Group Quarters

Our second data set is from a program of the University of Wisconsin Population Health Institute which is the County Health Rankings & Roadmaps. The data is a reflection on the health of a given county, as the measures included are physical and mental well-being that

are measured by the Health Outcomes of Length of Life and Quality of Life. This model is separated amongst Policies and Programs, Health Factors, and Health outcomes, and the 225 variables that fall in these for all of the Ohio Counties.



Variable Definitions:

- Dependent Variable: Primary Care Physician Rate
 - Primary Care Physician: Patients first point of contact
 - * Family Practice
 - * Internal Medicine
 - * Pediatrics
 - * OB/GYN
- Independent Variables:

- * Socioeconomic score | continuous variable on scale 0 to 1
- * Household Composition & Disability score | continuous variable on scale 0 to 1
- * Minority status and language | continuous variable on scale 0 to 1
- * Housing type & Transportation | continuous variable on scale 0 to 1

Exploratory Data Analysis

We will first begin with some exploratory data analysis to summarize SVI variables and health-care access measures across all the Ohio counties.

- We will first assess our dataset by creating a correlation matrix of our joined dataset to assess collinearity between our explanatory variables.
- We will then create visualizations for both data sets. We decided to create heat maps of different variables by county. These variables include some select variables we thought may be important and all four of the vulnerability indicators: Socioeconomic Status, Household Characteristics, Racial and Ethnic Minority Status, Housing type and Transportation. As well as, Primary Care Physician rate and Population to get a holistic look at the kinds of variables we would be comparing.
- We then create visuals after fitting the full model to assess its goodness of fit. These include the residual plots of the full model as well as a graph of predicted PCP rate on our full model versus the actual real PCP rate to give a better visual of our model's prediction accuracy.

Linear Regression Modeling

We fit the dataset to a linear regression model to quantify the relationship between variables that describe social vulnerability and Primary Care Physician rate.

- Full model: $Y_i = \beta_0 + \beta_1(SVI1) + \beta_2(SVI2) + \dots + \beta_{22}(SVI22) + \epsilon_i$
 - Let Y be the Primary Care Physician Rate
 - We fit the full model with all variables that were a part of the SVI dataset. This means for example no HS diploma, aged 17 or younger, english language proficiency, and more for a total of 22 explanatory variables.

Backwards Stepwise Regression

We did backwards stepwise regression on our full model to do variable selection. We based our model criteria on AIC values.

Software statement: All analyses will be conducted in R

Results

Instructions: Present the main numerical and graphical findings.

Overall the final model we came up with was:

We are choosing to focus on 0 and 0.001 significant codes. Which leaves us with 9 variables that have the most meaningful impact on Primary Care Physician Rate.

Significant code of 0:

- Below poverty
- No High school diploma
- Age 17 or under
- Housing Type transportation

Significant code of 0.001:

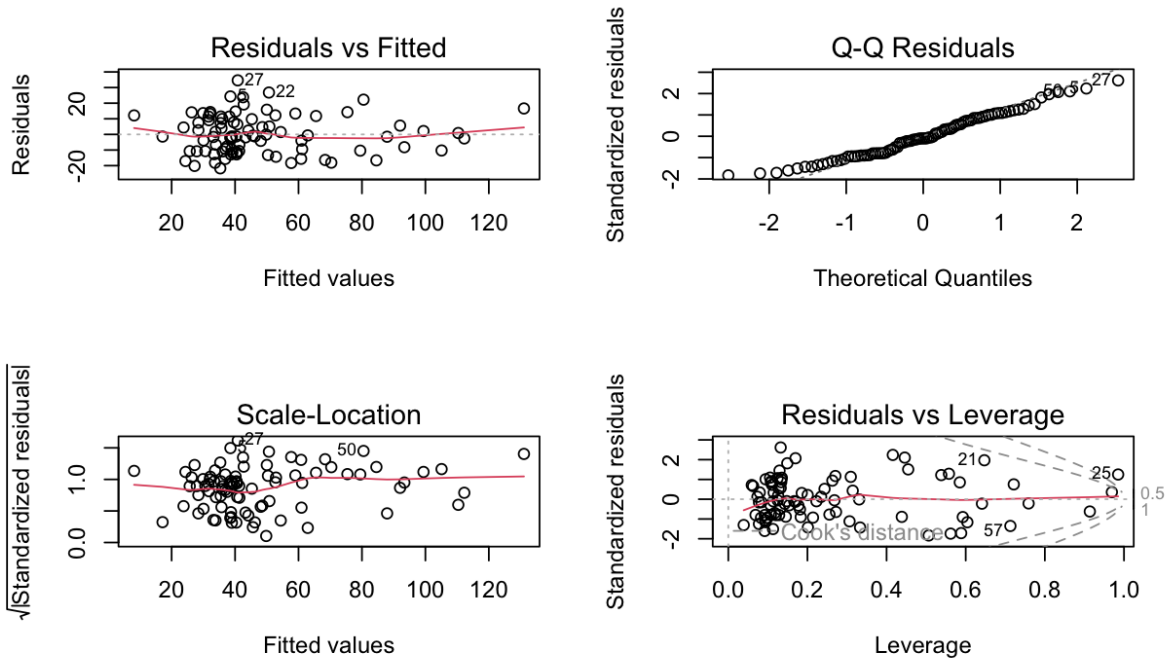
- Total population
- Occupied households
- Income
- Single parent household
- Crowding

Content to include:

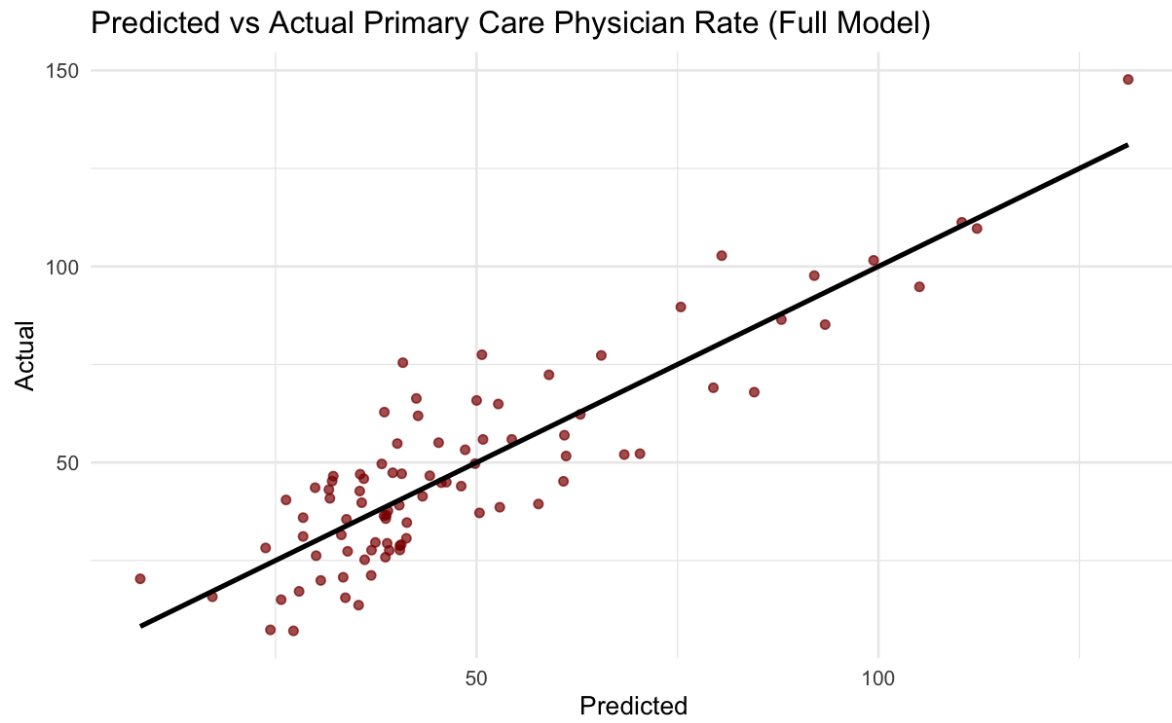
- Descriptive statistics summarizing the sample.
- Model results (e.g., regression coefficients, parameter estimates, classification accuracy).
- Figures and tables with clear labels and short explanations.

Important: Report results clearly, but do not evaluate their broader meaning here.

Full model visualizations



(residual plots)



(predicted vs actual PCP)

Call:

```
lm(formula = `Primary Care Physicians Rate` ~ total_population +  
housing_units + occupied_households + below_poverty + income +  
no_hs_diploma + age_17_or_under + single_parent_household +  
housing_type_transportation + multi_unit_structure + crowding,  
data = merged_data)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-24.207	-10.276	-1.199	9.152	34.792

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-6.314e+01	1.743e+01	-3.622	0.000525	***
total_population	-1.332e-03	4.877e-04	-2.731	0.007842	**
housing_units	-1.526e-03	1.110e-03	-1.375	0.173206	
occupied_households	3.714e-03	1.579e-03	2.353	0.021220	*
below_poverty	3.423e-03	9.042e-04	3.786	0.000304	***
income	3.349e-03	6.355e-04	5.269	1.24e-06	***
no_hs_diploma	-3.395e-03	9.823e-04	-3.456	0.000901	***
age_17_or_under	3.523e-03	1.121e-03	3.144	0.002380	**
single_parent_household	-7.543e-03	3.629e-03	-2.078	0.041050	*
housing_type_transportation	2.864e+01	6.861e+00	4.174	7.89e-05	***
multi_unit_structure	-1.071e-03	5.623e-04	-1.906	0.060491	.
crowding	-2.821e-02	6.601e-03	-4.273	5.52e-05	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 13.67 on 76 degrees of freedom

Multiple R-squared: 0.7546, Adjusted R-squared: 0.7191

F-statistic: 21.25 on 11 and 76 DF, p-value: < 2.2e-16

(Final Model After stepwise regression)

Conclusion

From looking at the final model we can see that below poverty, occupied households, income, age 17 or under, and housing type transportation all have a positive impact on the primary care physician rate. Meaning that for an increase of these variables, the expected primary care physician rate will increase. Then crowding, single parent household, total population, and no hs diploma all have a negative impact on the PCP rate. This means that as these variables

increase, the expected primary care physician rate decreases. These results do support our original hypothesis.

In the field of Public Health our statistical findings show that primary care physician rates are connected to different social determinants of health (SDOH) , and they influence the overall health and well-being of a given population. SDOH are important factors in public health that affect people's daily lives and therefore their health outcomes. Our findings point to specific variables of having a higher percentage below poverty and a large population of 17 and under linking to Primary Care Physician Rates. While having a higher population without a high school diploma being linked to a decrease in the Primary Care Physician rates. This means in the context of public health there are factors that professionals in this field can work to fix and see whether the outcome changes in these areas.

Our limitations include that the two datasets are from different years, the SVI data is from 2021 and the Ohio County data is from 2025. This might not be a full representation of the current variables that are affecting the primary care physician rate. The Ohio county data does not define what primary care physicians include. So in this case, we are assuming that this does not count the households that may use emergency rooms as their acute care instead of going to a primary care practice. There also might be other factors that may be impacting the primary care physician rate. The cross sectionality of the data makes it difficult to determine if the variables that we have chosen are the ones that are actually causing the effects. Not only could there be limitations within the dataset, but we are also only looking at county-level data. There could be inequality within the counties themselves.

Some things to do in the future to help increase the Primary Care Physician Rate includes strengthening education and health literacy programs in counties with a low high school diploma rate. This is because as we can see from the results, having a high number of people with no high school diploma has a negative effect on the primary care physician rate. Another way to help increase access through the PCP rate is by developing support services for single-parent households. By giving more support to these households, it can help decrease the number of people that are struggling with access.

Appendix

Here is an appendix of all our code.

```
# Imports
library(ggplot2)
library(dplyr)
```

```
Attaching package: 'dplyr'
```

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
v forcats   1.0.1      v stringr   1.6.0
```

```
v lubridate 1.9.4      v tibble    3.3.0
```

```
v purrr     1.2.0      v tidyr     1.3.1
```

```
v readr     2.1.5
```

```
-- Conflicts ----- tidyverse_conflicts() --
```

```
x dplyr::filter() masks stats::filter()
```

```
x dplyr::lag()     masks stats::lag()
```

```
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(readxl)
```

```
library(readr)
```

```
library(janitor)
```

Attaching package: 'janitor'

The following objects are masked from 'package:stats':

chisq.test, fisher.test

```
library(corrplot)
```

corrplot 0.95 loaded

```
library(car)
```

Loading required package: carData

Attaching package: 'car'

The following object is masked from 'package:purrr':

some

The following object is masked from 'package:dplyr':

recode

```
library(MASS)
```

Attaching package: 'MASS'

The following object is masked from 'package:dplyr':

select

```
library(ggrepel)
```

Setting up dataset

Import SVI dataset

```
# SVI Dataset Import
county_raw = read_csv("../data/SVI_County_Export.csv", show_col_types = FALSE) |>
  clean_names()

# SVI dataset cleaning
county_svi = county_raw |>
  separate(location, into = c("county_name", "state"), sep = ", ") |>
  mutate(
    county = county_name |>
      str_remove(" County") |>
      str_to_upper()
  )
prep = function(df) {
  df |>
```

```
mutate(across(where(is.character), as.factor)) |>
mutate(across(where(is.factor), ~fct_explicit_na(.x, na_level = "Missing"))) |>
mutate(across(where(is.numeric), ~ifelse(is.na(.x), median(.x, na.rm = TRUE), .x)))
}
```

```
county_svi_simple = prep(county_svi)
```

Warning: There was 1 warning in `mutate()`.

i In argument: `across(where(is.factor), ~fct\_explicit\_na(.x, na\_level = "Missing"))`.

Caused by warning:

! `fct\_explicit\_na()` was deprecated in forcats 1.0.0.

i Please use `fct\_na\_value\_to\_level()` instead.

```
county_svi_simple$county_name <- gsub(" County$", "", county_svi_simple$county_name)
county_svi_simple$fips <- sprintf("%05d", county_svi_simple$fips)
head(county_svi_simple)
```

```
# A tibble: 6 x 27
  county_name state fips x2018_overall_svi_score total_population housing_units
  <chr>      <fct> <chr>          <dbl>          <dbl>          <dbl>
1 Adams      Ohio  39001          0.931          27878          12920
2 Allen      Ohio  39003          0.770         103642          45063
3 Ashland    Ohio  39005          0.575          53477          22250
4 Ashtabula  Ohio  39007           1           98136          46107
5 Athens     Ohio  39009          0.942          65936          26610
6 Auglaize   Ohio  39011          0.023          45784          19852
# i 21 more variables: occupied_households <dbl>, socioeconomic_score <dbl>,
# below_poverty <dbl>, unemployed <dbl>, income <dbl>, no_hs_diploma <dbl>,
# house_composition_disability_score <dbl>, age_65_or_older <dbl>,
# age_17_or_under <dbl>, civilian_with_a_disability <dbl>,
# single_parent_household <dbl>, minority_status_language <dbl>,
# minority <dbl>, speak_english_less_than_well <dbl>,
# housing_type_transportation <dbl>, multi_unit_structure <dbl>, ...
```

Import Healthcare Access Dataset

```
# Healthcare access dataset
excel_sheets("../data/2025 County Health Rankings Ohio Data - v3.xlsx")
```

```
[1] "Introduction"           "Select Measure Data"
[3] "Select Measure Sources & Years" "Additional Measure Data"
[5] "Addtl Measure Sources & Years" "Health Groups"
```

```
main <- suppressMessages(read_excel("../data/2025 County Health Rankings Ohio Data - v3.xlsx",
                                   sheet = "Select Measure Data", skip = 1))
healthcare_vars <- c(
  "FIPS", "State", "County", # keep these for identification

  "# Primary Care Physicians",
  "Primary Care Physicians Rate",
  "Primary Care Physicians Ratio",
  "National Z-Score...90"
)

healthcare_access_data <- main[, healthcare_vars]
healthcare_access_data <- healthcare_access_data[-1, ]
```

Join datasets

```
#inner join the two datasets by county, wsanity checked 54 colns
merged_data = healthcare_access_data %>%
  left_join(county_svi_simple, by = c("FIPS" = "fips"))

head(merged_data)
```

```
# A tibble: 6 x 33
  FIPS State County `# Primary Care Physicians` Primary Care Physicians Ra~1
  <chr> <chr> <chr>                <dbl>                <dbl>
1 39001 Ohio Adams                12                43.6
2 39003 Ohio Allen                 66                64.9
3 39005 Ohio Ashland              23                44.0
4 39007 Ohio Ashtabula            28                28.8
5 39009 Ohio Athens               39                62.8
6 39011 Ohio Auglaize             16                34.7
# i abbreviated name: 1: `Primary Care Physicians Rate`
# i 28 more variables: `Primary Care Physicians Ratio` <chr>,
#   `National Z-Score...90` <dbl>, county_name <chr>, state <fct>,
#   x2018_overall_svi_score <dbl>, total_population <dbl>, housing_units <dbl>,
#   occupied_households <dbl>, socioeconomic_score <dbl>, below_poverty <dbl>,
#   unemployed <dbl>, income <dbl>, no_hs_diploma <dbl>,
#   house_composition_disability_score <dbl>, age_65_or_older <dbl>, ...
```

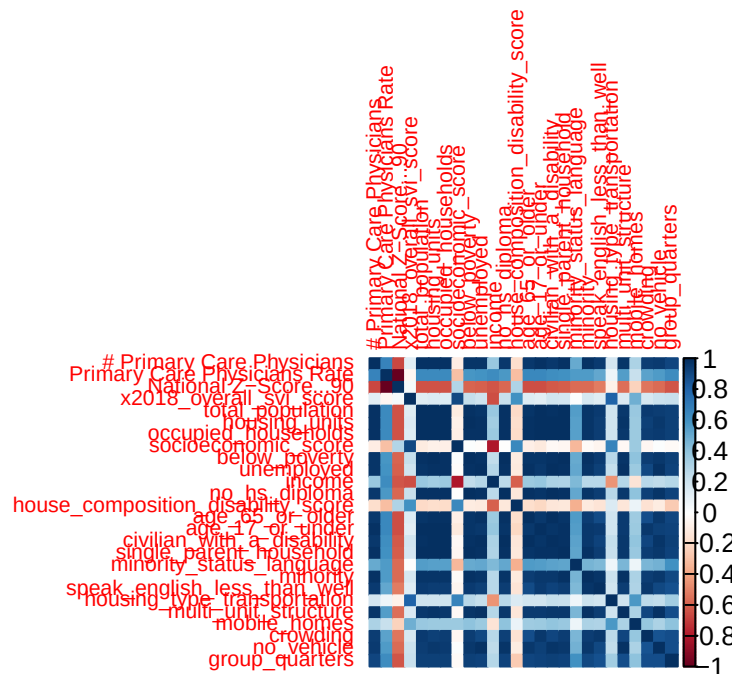
```
print(colnames(merged_data))
```

[1] "FIPS"	"State"
[3] "County"	"# Primary Care Physicians"
[5] "Primary Care Physicians Rate"	"Primary Care Physicians Ratio"
[7] "National Z-Score...90"	"county_name"
[9] "state"	"x2018_overall_svi_score"
[11] "total_population"	"housing_units"
[13] "occupied_households"	"socioeconomic_score"
[15] "below_poverty"	"unemployed"
[17] "income"	"no_hs_diploma"
[19] "house_composition_disability_score"	"age_65_or_older"
[21] "age_17_or_under"	"civilian_with_a_disability"
[23] "single_parent_household"	"minority_status_language"
[25] "minority"	"speak_english_less_than_well"
[27] "housing_type_transportation"	"multi_unit_structure"
[29] "mobile_homes"	"crowding"
[31] "no_vehicle"	"group_quarters"
[33] "county"	

EDA

Correlation matrix

```
numeric_vars <- dplyr::select(merged_data, where(is.numeric))
cor_matrix = cor(numeric_vars, use = "pairwise.complete.obs")
corrplot(cor_matrix, method = "color", tl.cex = 0.7)
```



Heat Maps

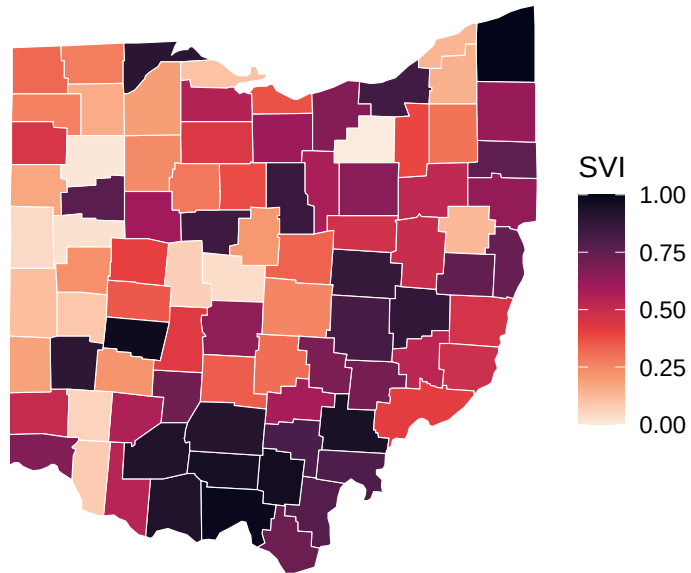
```
ohio_map <- map_data("county") %>%
  filter(region == "ohio") %>%
  mutate(county = toupper(subregion))

map_df <- ohio_map %>%
  left_join(merged_data, by = "county")

# Heat map: Ohio SVI score
ggplot(map_df, aes(long, lat, group = group, fill = x2018_overall_svi_score)) +
  geom_polygon(color = "white", size = 0.2) +
  coord_map() +
  scale_fill_viridis_c(option = "rocket", direction = -1) +
  theme_void() +
  labs(fill = "SVI", title = "Ohio Social Vulnerability Index (SVI)")
```

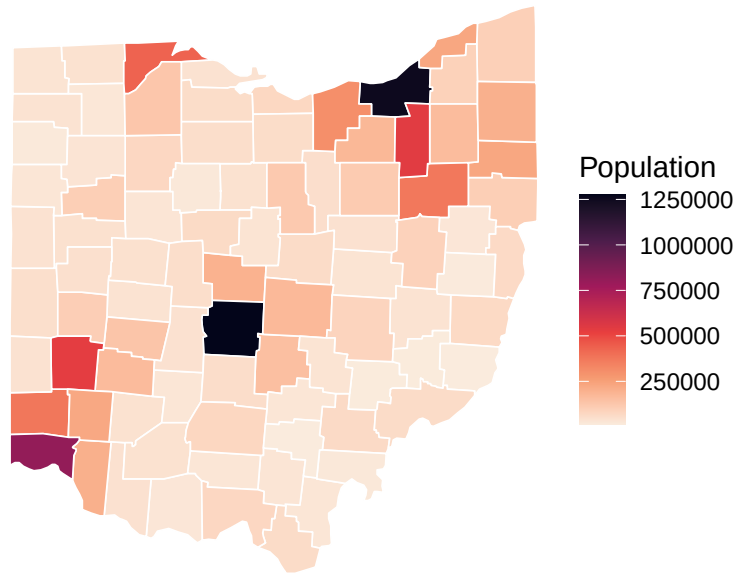
Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
i Please use `linewidth` instead.

Ohio Social Vulnerability Index (SVI)



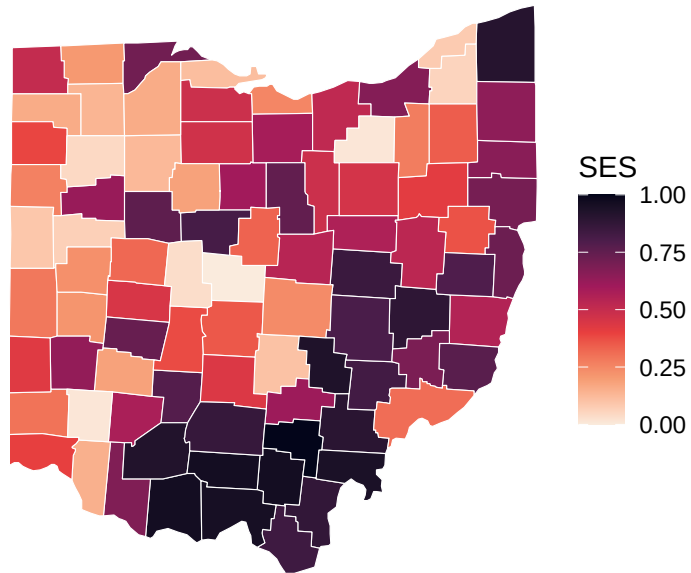
```
# Heat map: Total population
ggplot(map_df, aes(long, lat, group = group, fill = total_population)) +
  geom_polygon(color = "white", linewidth = 0.3) +
  coord_map() +
  scale_fill_viridis_c(option = "rocket", direction = -1) +
  theme_void() +
  labs(
    title = "Total Population by Ohio County",
    fill = "Population"
  )
```


Total Population by Ohio County



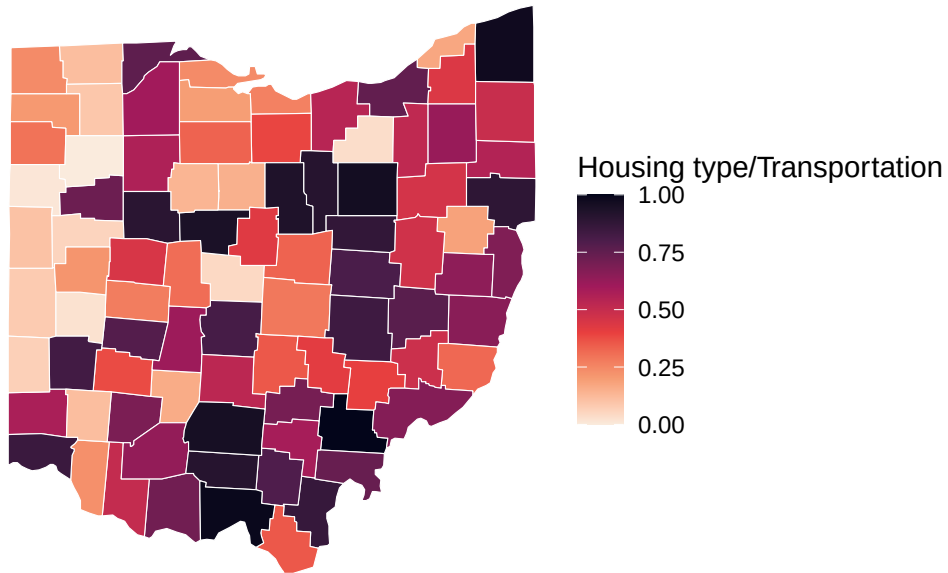
```
# Heat map: Socioeconomic Score
ggplot(map_df, aes(long, lat, group = group, fill = socioeconomic_score)) +
  geom_polygon(color = "white", size = 0.2) +
  coord_map() +
  scale_fill_viridis_c(option = "rocket", direction = -1) +
  theme_void() +
  labs(fill = "SES", title = "Socioeconomic score")
```

Socioeconomic score



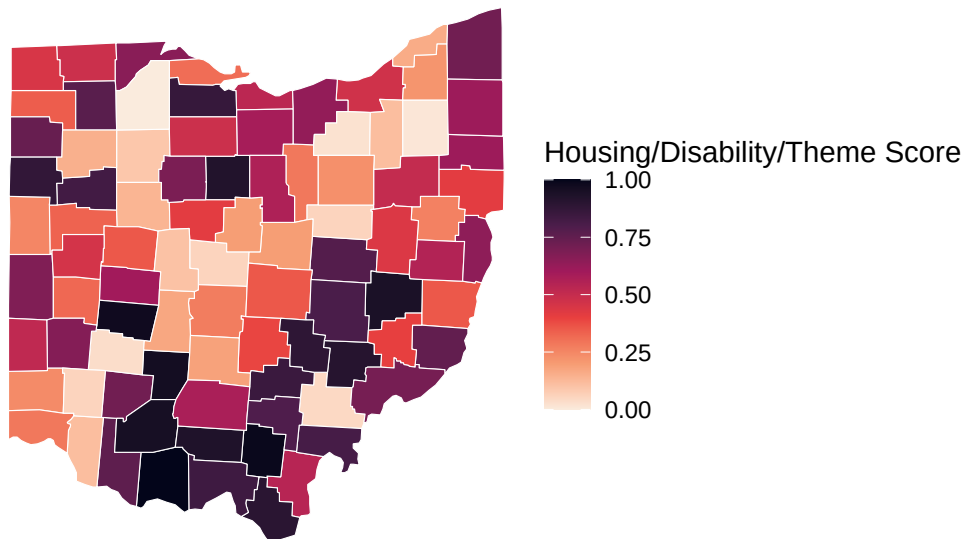
```
# Heat map: Housing Type & Transportation
ggplot(map_df, aes(long, lat, group = group, fill = housing_type_transportation)) +
  geom_polygon(color = "white", size = 0.2) +
  coord_map() +
  scale_fill_viridis_c(option = "rocket", direction = -1) +
  theme_void() +
  labs(fill = "Housing type/Transportation", title = "Housing Type & Transportation")
```

Housing Type & Transportation



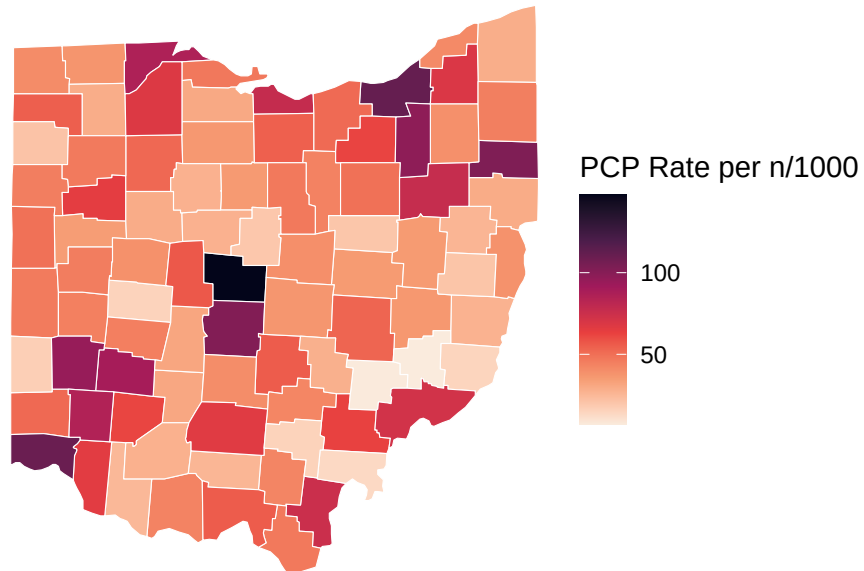
```
# Heat map: Household Composition & Disability Vulnerability
ggplot(map_df, aes(long, lat, group = group, fill = house_composition_disability_score)) +
  geom_polygon(color = "white", linewidth = 0.2) +
  coord_map() +
  scale_fill_viridis_c(option = "rocket", direction = -1) +
  theme_void() +
  labs(
    title = "Household Composition & Disability Vulnerability\nby Ohio County",
    fill = "Housing/Disability/Theme Score",
  )
```

Household Composition & Disability Vulnerability by Ohio County



```
# Heat map: Primary Care Physician Rate
ggplot(map_df, aes(long, lat, group = group, fill = `Primary Care Physicians Rate`)) +
  geom_polygon(color = "white", linewidth = 0.2) +
  coord_map() +
  scale_fill_viridis_c(option = "rocket", direction = -1) +
  theme_void() +
  labs(
    title = "Primary Care Physician Rate by Ohio County",
    fill = "PCP Rate per n/1000",
  )
```

Primary Care Physician Rate by Ohio County



Linear Model

```
full_mod = lm(  
  `Primary Care Physicians Rate` ~  
    total_population +  
    housing_units +  
    occupied_households +  
    socioeconomic_score +  
    below_poverty +  
    unemployed +  
    income +  
    no_hs_diploma +  
    house_composition_disability_score +  
    age_65_or_older +  
    age_17_or_under +  
    civilian_with_a_disability +  
    single_parent_household +  
    minority_status_language +  
    minority +  
    speak_english_less_than_well +  
    housing_type_transportation+  
    multi_unit_structure +
```

```

mobile_homes +
crowding +
no_vehicle +
group_quarters,
data = merged_data
)
summary(full_mod)

```

Call:

```

lm(formula = `Primary Care Physicians Rate` ~ total_population +
housing_units + occupied_households + socioeconomic_score +
below_poverty + unemployed + income + no_hs_diploma + house_composition_disability_score +
age_65_or_older + age_17_or_under + civilian_with_a_disability +
single_parent_household + minority_status_language + minority +
speak_english_less_than_well + housing_type_transportation +
multi_unit_structure + mobile_homes + crowding + no_vehicle +
group_quarters, data = merged_data)

```

Residuals:

	Min	1Q	Median	3Q	Max
	-21.729	-10.641	-1.293	9.791	34.610

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-5.197e+01	2.909e+01	-1.786	0.07873 .
total_population	-2.217e-03	1.058e-03	-2.096	0.04001 *
housing_units	-2.032e-03	1.400e-03	-1.451	0.15148
occupied_households	5.681e-03	2.332e-03	2.436	0.01760 *
socioeconomic_score	-1.588e+01	1.564e+01	-1.015	0.31374
below_poverty	3.547e-03	1.190e-03	2.981	0.00404 **
unemployed	-9.133e-04	4.953e-03	-0.184	0.85427
income	2.751e-03	9.569e-04	2.875	0.00546 **
no_hs_diploma	-4.229e-03	1.898e-03	-2.228	0.02938 *
house_composition_disability_score	1.052e+01	1.082e+01	0.973	0.33436
age_65_or_older	-1.278e-04	1.763e-03	-0.072	0.94245
age_17_or_under	5.141e-03	2.186e-03	2.352	0.02172 *
civilian_with_a_disability	1.517e-03	1.464e-03	1.036	0.30402
single_parent_household	-1.317e-02	6.047e-03	-2.178	0.03303 *
minority_status_language	8.571e+00	8.346e+00	1.027	0.30821
minority	2.430e-04	7.115e-04	0.342	0.73379
speak_english_less_than_well	1.046e-03	6.749e-03	0.155	0.87731

housing_type_transportation	3.392e+01	1.186e+01	2.860	0.00570	**
multi_unit_structure	-1.365e-03	1.536e-03	-0.889	0.37731	
mobile_homes	8.254e-04	2.803e-03	0.295	0.76931	
crowding	-2.996e-02	1.153e-02	-2.598	0.01158	*
no_vehicle	-4.401e-04	3.117e-03	-0.141	0.88817	
group_quarters	7.670e-04	1.615e-03	0.475	0.63640	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 14.2 on 65 degrees of freedom

Multiple R-squared: 0.7736, Adjusted R-squared: 0.6969

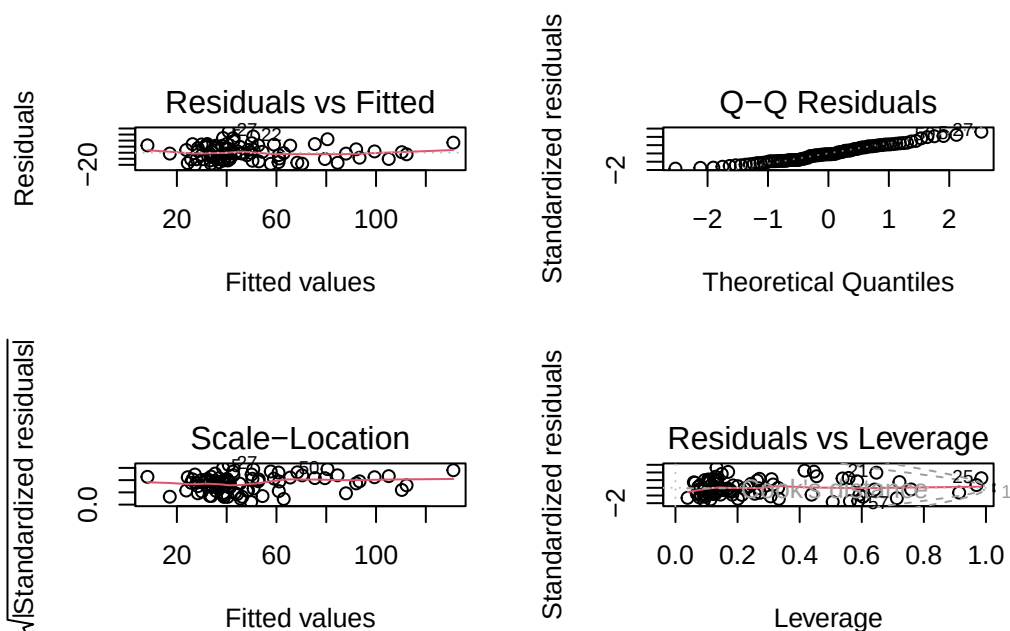
F-statistic: 10.09 on 22 and 65 DF, p-value: 1.533e-13

Full Model Visualizations

```
# Residuals plots
par(mfrow = c(2,2))
plot(full_mod)
```

Warning in sqrt(crit * p * (1 - hh)/hh): NaNs produced

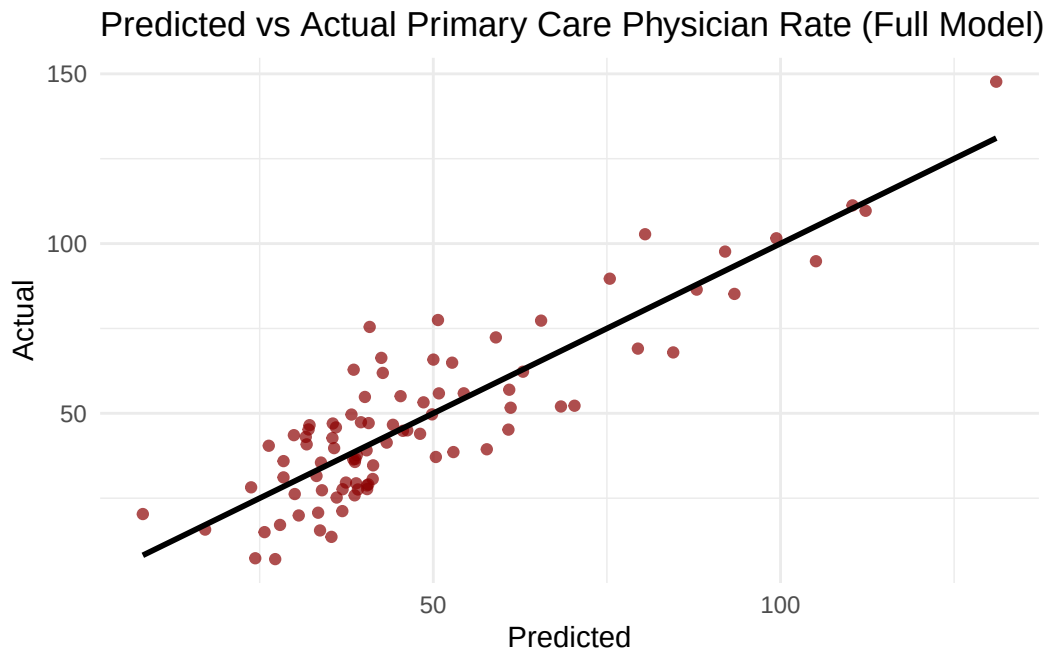
Warning in sqrt(crit * p * (1 - hh)/hh): NaNs produced



```
# PCP Prediction plot
merged_data$pred_full <- predict(full_mod)

ggplot(merged_data,
       aes(x = pred_full, y = `Primary Care Physicians Rate`)) +
  geom_point(alpha = 0.7, color = "darkred") +
  geom_smooth(method = "lm", color = "black", se = FALSE) +
  labs(
    title = "Predicted vs Actual Primary Care Physician Rate (Full Model)",
    x = "Predicted",
    y = "Actual"
  ) +
  theme_minimal()
```

`geom\_smooth()` using formula = 'y ~ x'



Backwards Stepwise Regression

```
step_mod = stepAIC(full_mod, direction = "backward")
```


Start: AIC=486.36

```
`Primary Care Physicians Rate` ~ total_population + housing_units +
  occupied_households + socioeconomic_score + below_poverty +
  unemployed + income + no_hs_diploma + house_composition_disability_score +
  age_65_or_older + age_17_or_under + civilian_with_a_disability +
  single_parent_household + minority_status_language + minority +
  speak_english_less_than_well + housing_type_transportation +
  multi_unit_structure + mobile_homes + crowding + no_vehicle +
  group_quarters
```

	Df	Sum of Sq	RSS	AIC
- age_65_or_older	1	1.06	13115	484.37
- no_vehicle	1	4.02	13118	484.39
- speak_english_less_than_well	1	4.85	13119	484.39
- unemployed	1	6.86	13121	484.41
- mobile_homes	1	17.50	13131	484.48
- minority	1	23.54	13137	484.52
- group_quarters	1	45.51	13159	484.66
- multi_unit_structure	1	159.43	13273	485.42
- house_composition_disability_score	1	190.85	13305	485.63
- socioeconomic_score	1	207.96	13322	485.74
- minority_status_language	1	212.81	13327	485.78
- civilian_with_a_disability	1	216.56	13330	485.80
<none>			13114	486.36
- housing_units	1	425.01	13539	487.17
- total_population	1	886.03	14000	490.11
- single_parent_household	1	957.17	14071	490.56
- no_hs_diploma	1	1001.10	14115	490.83
- age_17_or_under	1	1116.01	14230	491.55
- occupied_households	1	1197.28	14311	492.05
- crowding	1	1362.01	14476	493.06
- housing_type_transportation	1	1649.74	14764	494.79
- income	1	1667.58	14782	494.89
- below_poverty	1	1793.16	14907	495.64

Step: AIC=484.37

```
`Primary Care Physicians Rate` ~ total_population + housing_units +
  occupied_households + socioeconomic_score + below_poverty +
  unemployed + income + no_hs_diploma + house_composition_disability_score +
  age_17_or_under + civilian_with_a_disability + single_parent_household +
  minority_status_language + minority + speak_english_less_than_well +
  housing_type_transportation + multi_unit_structure + mobile_homes +
  crowding + no_vehicle + group_quarters
```

	Df	Sum of Sq	RSS	AIC
- no_vehicle	1	5.14	13120	482.40
- speak_english_less_than_well	1	5.64	13121	482.40
- unemployed	1	7.94	13123	482.42
- mobile_homes	1	26.32	13141	482.54
- minority	1	27.26	13142	482.55
- group_quarters	1	46.26	13161	482.68
- multi_unit_structure	1	167.95	13283	483.49
- house_composition_disability_score	1	196.28	13311	483.67
- socioeconomic_score	1	207.74	13323	483.75
- minority_status_language	1	211.75	13327	483.78
- civilian_with_a_disability	1	228.68	13344	483.89
<none>			13115	484.37
- housing_units	1	498.20	13613	485.65
- total_population	1	886.34	14001	488.12
- single_parent_household	1	1052.40	14167	489.16
- age_17_or_under	1	1194.86	14310	490.04
- occupied_households	1	1243.88	14359	490.34
- no_hs_diploma	1	1250.09	14365	490.38
- crowding	1	1411.46	14526	491.36
- housing_type_transportation	1	1648.71	14764	492.79
- income	1	1724.32	14839	493.24
- below_poverty	1	1815.14	14930	493.77

Step: AIC=482.4

```
`Primary Care Physicians Rate` ~ total_population + housing_units +
  occupied_households + socioeconomic_score + below_poverty +
  unemployed + income + no_hs_diploma + house_composition_disability_score +
  age_17_or_under + civilian_with_a_disability + single_parent_household +
  minority_status_language + minority + speak_english_less_than_well +
  housing_type_transportation + multi_unit_structure + mobile_homes +
  crowding + group_quarters
```

	Df	Sum of Sq	RSS	AIC
- unemployed	1	13.26	13133	480.49
- speak_english_less_than_well	1	16.28	13136	480.51
- minority	1	22.51	13143	480.55
- mobile_homes	1	38.13	13158	480.66
- group_quarters	1	48.85	13169	480.73
- house_composition_disability_score	1	194.37	13314	481.70
- multi_unit_structure	1	195.77	13316	481.70
- socioeconomic_score	1	202.60	13323	481.75

- minority_status_language	1	207.62	13328	481.78
- civilian_with_a_disability	1	236.57	13357	481.97
<none>			13120	482.40
- housing_units	1	517.57	13638	483.81
- total_population	1	971.99	14092	486.69
- age_17_or_under	1	1207.54	14328	488.15
- occupied_households	1	1351.19	14471	489.03
- single_parent_household	1	1355.59	14476	489.05
- crowding	1	1438.36	14558	489.56
- housing_type_transportation	1	1754.31	14874	491.45
- no_hs_diploma	1	1785.87	14906	491.63
- income	1	1786.68	14907	491.64
- below_poverty	1	1828.20	14948	491.88

Step: AIC=480.49

`Primary Care Physicians Rate` ~ total_population + housing_units +
 occupied_households + socioeconomic_score + below_poverty +
 income + no_hs_diploma + house_composition_disability_score +
 age_17_or_under + civilian_with_a_disability + single_parent_household +
 minority_status_language + minority + speak_english_less_than_well +
 housing_type_transportation + multi_unit_structure + mobile_homes +
 crowding + group_quarters

	Df	Sum of Sq	RSS	AIC
- speak_english_less_than_well	1	16.27	13150	478.60
- minority	1	19.22	13153	478.62
- group_quarters	1	50.29	13184	478.83
- mobile_homes	1	53.39	13187	478.85
- house_composition_disability_score	1	189.64	13323	479.75
- socioeconomic_score	1	198.82	13332	479.81
- minority_status_language	1	215.65	13349	479.92
- civilian_with_a_disability	1	226.53	13360	480.00
- multi_unit_structure	1	237.28	13371	480.07
<none>			13133	480.49
- housing_units	1	561.30	13695	482.17
- total_population	1	1041.75	14175	485.21
- age_17_or_under	1	1279.36	14413	486.67
- single_parent_household	1	1376.95	14510	487.26
- occupied_households	1	1471.66	14605	487.84
- crowding	1	1671.57	14805	489.03
- housing_type_transportation	1	1752.21	14886	489.51
- income	1	1778.39	14912	489.67
- no_hs_diploma	1	1832.88	14966	489.99

- below_poverty 1 1878.57 15012 490.26

Step: AIC=478.6

`Primary Care Physicians Rate` ~ total_population + housing_units +
 occupied_households + socioeconomic_score + below_poverty +
 income + no_hs_diploma + house_composition_disability_score +
 age_17_or_under + civilian_with_a_disability + single_parent_household +
 minority_status_language + minority + housing_type_transportation +
 multi_unit_structure + mobile_homes + crowding + group_quarters

	Df	Sum of Sq	RSS	AIC
- group_quarters	1	34.05	13184	476.83
- mobile_homes	1	63.96	13214	477.03
- house_composition_disability_score	1	200.88	13350	477.93
- civilian_with_a_disability	1	210.51	13360	478.00
- socioeconomic_score	1	224.39	13374	478.09
- minority_status_language	1	238.39	13388	478.18
- minority	1	264.02	13414	478.35
<none>			13150	478.60
- multi_unit_structure	1	609.66	13759	480.59
- housing_units	1	617.99	13768	480.64
- occupied_households	1	1463.65	14613	485.89
- single_parent_household	1	1579.51	14729	486.58
- total_population	1	1646.51	14796	486.98
- housing_type_transportation	1	1744.38	14894	487.56
- crowding	1	1828.80	14978	488.06
- income	1	1830.17	14980	488.07
- below_poverty	1	1899.03	15049	488.47
- no_hs_diploma	1	2004.04	15154	489.08
- age_17_or_under	1	2211.05	15361	490.28

Step: AIC=476.83

`Primary Care Physicians Rate` ~ total_population + housing_units +
 occupied_households + socioeconomic_score + below_poverty +
 income + no_hs_diploma + house_composition_disability_score +
 age_17_or_under + civilian_with_a_disability + single_parent_household +
 minority_status_language + minority + housing_type_transportation +
 multi_unit_structure + mobile_homes + crowding

	Df	Sum of Sq	RSS	AIC
- mobile_homes	1	47.34	13231	475.14
- house_composition_disability_score	1	168.10	13352	475.94
- socioeconomic_score	1	216.57	13400	476.26

- civilian_with_a_disability	1	241.93	13426	476.43
- minority	1	268.26	13452	476.60
- minority_status_language	1	276.68	13460	476.65
<none>			13184	476.83
- housing_units	1	599.79	13784	478.74
- multi_unit_structure	1	621.04	13805	478.88
- occupied_households	1	1465.55	14649	484.10
- single_parent_household	1	1545.79	14730	484.58
- total_population	1	1739.23	14923	485.73
- income	1	1821.85	15006	486.22
- crowding	1	1866.16	15050	486.48
- below_poverty	1	1978.11	15162	487.13
- housing_type_transportation	1	2222.76	15406	488.54
- age_17_or_under	1	2226.88	15410	488.56
- no_hs_diploma	1	2338.85	15522	489.20

Step: AIC=475.14

```
`Primary Care Physicians Rate` ~ total_population + housing_units +
  occupied_households + socioeconomic_score + below_poverty +
  income + no_hs_diploma + house_composition_disability_score +
  age_17_or_under + civilian_with_a_disability + single_parent_household +
  minority_status_language + minority + housing_type_transportation +
  multi_unit_structure + crowding
```

	Df	Sum of Sq	RSS	AIC
- house_composition_disability_score	1	216.86	13448	474.57
- socioeconomic_score	1	218.86	13450	474.59
- minority	1	231.02	13462	474.67
- minority_status_language	1	234.99	13466	474.69
- civilian_with_a_disability	1	250.84	13482	474.80
<none>			13231	475.14
- housing_units	1	562.07	13793	476.80
- multi_unit_structure	1	575.18	13806	476.89
- occupied_households	1	1438.03	14669	482.22
- single_parent_household	1	1623.08	14854	483.33
- total_population	1	1700.20	14931	483.78
- crowding	1	1819.74	15051	484.48
- income	1	1959.24	15190	485.29
- age_17_or_under	1	2199.96	15431	486.68
- no_hs_diploma	1	2381.09	15612	487.70
- housing_type_transportation	1	2522.31	15753	488.50
- below_poverty	1	2649.10	15880	489.20

Step: AIC=474.57

```
`Primary Care Physicians Rate` ~ total_population + housing_units +
  occupied_households + socioeconomic_score + below_poverty +
  income + no_hs_diploma + age_17_or_under + civilian_with_a_disability +
  single_parent_household + minority_status_language + minority +
  housing_type_transportation + multi_unit_structure + crowding
```

	Df	Sum of Sq	RSS	AIC
- socioeconomic_score	1	66.27	13514	473.01
- minority_status_language	1	239.16	13687	474.12
- minority	1	287.07	13735	474.43
<none>			13448	474.57
- civilian_with_a_disability	1	431.02	13879	475.35
- housing_units	1	589.13	14037	476.35
- multi_unit_structure	1	689.81	14138	476.98
- single_parent_household	1	1476.53	14924	481.74
- occupied_households	1	1566.88	15015	482.27
- crowding	1	1629.72	15078	482.64
- income	1	1941.06	15389	484.44
- total_population	1	2077.53	15525	485.21
- housing_type_transportation	1	2348.57	15796	486.74
- age_17_or_under	1	2387.63	15836	486.96
- below_poverty	1	2446.83	15895	487.28
- no_hs_diploma	1	2543.88	15992	487.82

Step: AIC=473.01

```
`Primary Care Physicians Rate` ~ total_population + housing_units +
  occupied_households + below_poverty + income + no_hs_diploma +
  age_17_or_under + civilian_with_a_disability + single_parent_household +
  minority_status_language + minority + housing_type_transportation +
  multi_unit_structure + crowding
```

	Df	Sum of Sq	RSS	AIC
- minority	1	243.5	13758	472.58
<none>			13514	473.01
- minority_status_language	1	330.1	13844	473.13
- civilian_with_a_disability	1	389.4	13904	473.51
- housing_units	1	553.6	14068	474.54
- multi_unit_structure	1	635.2	14149	475.05
- single_parent_household	1	1411.4	14926	479.75
- occupied_households	1	1502.4	15016	480.28
- crowding	1	1597.6	15112	480.84
- total_population	1	2028.4	15542	483.31

- below_poverty	1	2390.5	15905	485.34
- age_17_or_under	1	2417.7	15932	485.49
- no_hs_diploma	1	2479.1	15993	485.83
- housing_type_transportation	1	2858.6	16373	487.89
- income	1	3920.5	17435	493.42

Step: AIC=472.58

```
`Primary Care Physicians Rate` ~ total_population + housing_units +
  occupied_households + below_poverty + income + no_hs_diploma +
  age_17_or_under + civilian_with_a_disability + single_parent_household +
  minority_status_language + housing_type_transportation +
  multi_unit_structure + crowding
```

	Df	Sum of Sq	RSS	AIC
- civilian_with_a_disability	1	249.2	14007	472.16
- minority_status_language	1	292.4	14050	472.43
<none>			13758	472.58
- housing_units	1	419.8	14177	473.22
- multi_unit_structure	1	420.6	14178	473.23
- single_parent_household	1	1179.2	14937	477.81
- occupied_households	1	1262.8	15020	478.31
- crowding	1	1465.0	15223	479.48
- total_population	1	1797.2	15555	481.38
- age_17_or_under	1	2242.0	16000	483.86
- no_hs_diploma	1	2313.0	16071	484.25
- housing_type_transportation	1	2622.6	16380	485.93
- below_poverty	1	2735.0	16493	486.53
- income	1	4121.3	17879	493.64

Step: AIC=472.16

```
`Primary Care Physicians Rate` ~ total_population + housing_units +
  occupied_households + below_poverty + income + no_hs_diploma +
  age_17_or_under + single_parent_household + minority_status_language +
  housing_type_transportation + multi_unit_structure + crowding
```

	Df	Sum of Sq	RSS	AIC
- minority_status_language	1	205.4	14212	471.44
- housing_units	1	285.6	14292	471.93
<none>			14007	472.16
- multi_unit_structure	1	619.2	14626	473.96
- single_parent_household	1	944.0	14951	475.90
- occupied_households	1	1045.5	15052	476.49
- total_population	1	1552.4	15559	479.41

- age_17_or_under	1	1999.1	16006	481.90
- no_hs_diploma	1	2307.4	16314	483.58
- below_poverty	1	2738.8	16746	485.87
- housing_type_transportation	1	2911.5	16918	486.78
- crowding	1	3009.7	17016	487.29
- income	1	4367.2	18374	494.04

Step: AIC=471.44

```
`Primary Care Physicians Rate` ~ total_population + housing_units +
  occupied_households + below_poverty + income + no_hs_diploma +
  age_17_or_under + single_parent_household + housing_type_transportation +
  multi_unit_structure + crowding
```

	Df	Sum of Sq	RSS	AIC
<none>			14212	471.44
- housing_units	1	353.5	14566	471.60
- multi_unit_structure	1	679.0	14891	473.54
- single_parent_household	1	807.8	15020	474.30
- occupied_households	1	1035.2	15247	475.62
- total_population	1	1394.9	15607	477.68
- age_17_or_under	1	1848.1	16060	480.20
- no_hs_diploma	1	2233.3	16446	482.28
- below_poverty	1	2680.3	16892	484.64
- housing_type_transportation	1	3258.0	17470	487.60
- crowding	1	3414.6	17627	488.39
- income	1	5192.3	19404	496.84

Final Model

```
summary(step_mod)
```

Call:

```
lm(formula = `Primary Care Physicians Rate` ~ total_population +
  housing_units + occupied_households + below_poverty + income +
  no_hs_diploma + age_17_or_under + single_parent_household +
  housing_type_transportation + multi_unit_structure + crowding,
  data = merged_data)
```

Residuals:

Min	1Q	Median	3Q	Max
-----	----	--------	----	-----

-24.207 -10.276 -1.199 9.152 34.792

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-6.314e+01	1.743e+01	-3.622	0.000525	***
total_population	-1.332e-03	4.877e-04	-2.731	0.007842	**
housing_units	-1.526e-03	1.110e-03	-1.375	0.173206	
occupied_households	3.714e-03	1.579e-03	2.353	0.021220	*
below_poverty	3.423e-03	9.042e-04	3.786	0.000304	***
income	3.349e-03	6.355e-04	5.269	1.24e-06	***
no_hs_diploma	-3.395e-03	9.823e-04	-3.456	0.000901	***
age_17_or_under	3.523e-03	1.121e-03	3.144	0.002380	**
single_parent_household	-7.543e-03	3.629e-03	-2.078	0.041050	*
housing_type_transportation	2.864e+01	6.861e+00	4.174	7.89e-05	***
multi_unit_structure	-1.071e-03	5.623e-04	-1.906	0.060491	.
crowding	-2.821e-02	6.601e-03	-4.273	5.52e-05	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 13.67 on 76 degrees of freedom

Multiple R-squared: 0.7546, Adjusted R-squared: 0.7191

F-statistic: 21.25 on 11 and 76 DF, p-value: < 2.2e-16